

WASP-36b

R-2035

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_140409 · created 2026-08-08T01:41:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456756.6858 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.048
Transit depth (Rp/Rs)^2	2.23 +/- 1.43 [%]
Orbital Inclination (inc)	80.74 +/- 2.9
Airmass coefficient 1 (a1)	0.137 +/- 0.086
Airmass coefficient 2 (a2)	1.25 +/- 0.5
Scatter in the residuals of the lightcurve fit is	63.4 %
Best Comparison Star	#1 - [560, 42]
Optimal Aperture	0.99
Optimal Annulus	3.0
Transit Duration (day)	0.047 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.048 vs 0.1368 (deviation 0.2σ)
Duration fit vs published	0.047 d vs 0.0758 d (deviation 0.69σ)
Mid-time O–C	2.4 min
Depth SNR	2.4
Residual scatter	63.4 %

Light curve

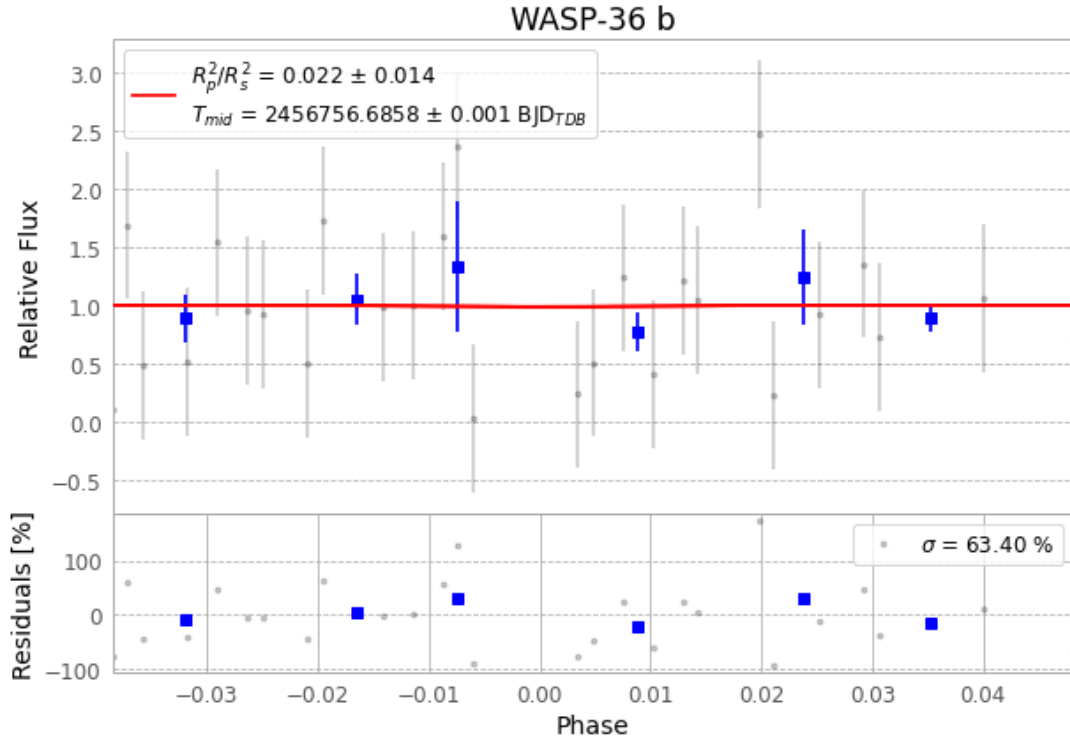


Figure 1 — FinalLightCurve_WASP-36 b_2014-04-08.png (SHA-256 75c440db5017454f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [515, 383], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 03c32ab080f54517...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

68 FITS frames (45 MB), WASP-36140409025713.FITS → WASP-36140409061812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-140409.log (d2a87133cbcc...), FinalLightCurve_WASP-36 b_2014-04-08.png (75c440db5017...), FinalLightCurve_WASP-36 b_2014-04-08.csv (cf697f08058d...)

Compute resources

Wall time 133.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 01:41Z.